



ENGINEERING — INTERNAL

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FLYN AI — Smart Agent Addons

Full specification: roles, training data, system prompts & implementation

Document: AI Agent Addons — Full Specification

For: FLYN AI Engineering Team — Internal

What This Document Covers

This specification defines four plug-and-play AI Agent Addons for the FLYN AI platform. Each addon is a pre-built, pre-trained autonomous AI agent that a company activates from their FLYN dashboard with zero workflow setup. The moment it is switched on, the agent begins operating independently — handling the full scope of a human role in that function. Each agent is trained with deep domain knowledge, a company-specific onboarding layer, and a full suite of automated actions. For each agent this document defines: the human role it replaces, every function it performs, the system prompt and training instructions that power its intelligence, the onboarding data it collects from the company, the triggers and automations it runs, and the implementation specification for the engineering team.

AGENT	HUMAN ROLE EQUIVALENT	PRIMARY CHANNEL	KEY OUTCOME
AI Marketing Agent	Marketing Manager + Campaign Executive	WhatsApp / Telegram	Generates and converts leads automatically
AI Content Creator	Copywriter + Content Strategist	WhatsApp / Telegram / Web	Produces on-brand content on demand and on schedule
AI Social Media Manager	Social Media Manager + Community Manager	WhatsApp / Telegram + social APIs	Posts, engages, grows audience 24/7
AI Front Desk / Support	Receptionist + Customer Support Agent	WhatsApp / Telegram	Answers, qualifies, books, resolves — always on

Shared Architecture — How All Agents Work

Every agent is implemented as a pre-configured flow in the FLYN flow engine with a master system prompt stored in the `agent_configs` table. On activation, the agent subscribes to inbound message triggers and scheduled cron triggers for its channel. It uses the FLYN AI Chat node (OpenAI / Gemini / DeepSeek) as its reasoning core. Company-specific data (name, tone, products, FAQs, etc.) collected during the onboarding wizard is injected into the system prompt as a `COMPANY CONTEXT` block. No manual flow building is required by the end user — the agent is entirely self-contained.

Onboarding Wizard — Implementation Note (applies to all agents)

Each agent triggers a 5-10 message onboarding conversation on first activation. Responses are stored in the agent_company_data table (agent_id, user_id, field_key, field_value). After completion, the system assembles the COMPANY CONTEXT block and inserts it into the agent's master system prompt. The agent is then live. Users can re-run onboarding at any time from Settings.

1.1 Human Role This Agent Replaces

In a human organisation, the marketing function is staffed by a Marketing Manager who sets strategy, a Campaign Executive who writes and sends campaigns, a Lead Generation Specialist who builds contact lists and runs outreach, a CRM executive who follows up and nurtures leads, and a Sales Coordinator who qualifies and hands off ready leads to the sales team. This agent performs all of these roles simultaneously, around the clock, on WhatsApp and Telegram.

1.2 Full Function List

Function	What the Agent Does Autonomously	Trigger
Lead capture	Receives inbound enquiries on WhatsApp/Telegram, extracts contact details, intent, and budget. Creates a contact record and adds to the leads pipeline.	Inbound message
Lead qualification	Asks a structured set of qualifying questions (budget, timeline, need, authority). Scores the lead and tags it (Hot / Warm / Cold) in the CRM.	After lead capture
Outbound prospecting	Sends personalised first-touch WhatsApp messages to new contact lists uploaded by the user. Personalises each message using the contact's name and list segment.	New list uploaded or scheduled
Drip campaign management	Runs multi-step follow-up sequences. Sends message 1 immediately, message 2 after 24 hrs if no reply, message 3 after 48 hrs, etc. Stops when the lead replies.	Lead enters pipeline
Promotional broadcasts	Composes and sends promotional campaigns to segmented lists on a scheduled basis (weekly deals, flash sales, seasonal offers).	Scheduled cron / manual trigger
Re-engagement campaigns	Identifies contacts who have not responded in 14+ days and sends a re-engagement message with a new angle or offer.	Scheduled cron — daily check
Upsell / cross-sell	After a purchase webhook, sends a personalised upsell or complementary product message based on what the customer bought.	WooCommerce/Shopify order webhook
Referral requests	After a positive interaction or purchase, sends a referral request asking the customer to share a link or refer a friend.	3 days after purchase webhook
Campaign performance reporting	Sends a weekly WhatsApp summary to the business owner: messages sent, replies received, leads qualified, conversions, and top-performing campaign.	Scheduled — every Monday 9am

Objection handling	When a lead raises a price, timing, or trust objection, the agent identifies it and responds with a pre-trained objection-handling script tailored to the company.	Any inbound message with objection keywords
Appointment / demo booking	When a lead is ready, the agent offers available slots and confirms a booking via a calendar link or by collecting preferred time and notifying the human team.	Hot lead signal

1.3 Onboarding Data Collected from Company

Data Field	Why the Agent Needs It
Business name & industry	Used in every outbound message and campaign for personalisation
Products / services offered (list each with price range)	Powers product-specific campaign content and upsell logic
Target customer profile (who you sell to)	Shapes qualifying questions and prospect outreach tone
Unique selling points (what makes you better)	Inserted into objection handling and first-touch messages
Current promotions or offers	Used in broadcast and prospecting messages
Brand tone of voice (formal / friendly / bold)	Applied to all AI-generated message copy
Qualifying questions (budget, timeline, need, authority)	Sets the lead qualification flow
Calendar / booking link	Used when a hot lead is ready to book
Objections you commonly face	Trains the objection-handling responses
WhatsApp number / Telegram bot for notifications	Where the weekly report is sent

1.4 Master System Prompt (stored in agent_configs)

You are the AI Marketing Agent for {{company_name}}, a {{industry}} business.

Your job is to operate as a fully autonomous marketing department – generating leads, running campaigns, qualifying prospects, handling objections, and converting interest into booked appointments or confirmed sales. You work 24 hours a day, 7 days a week.

COMPANY CONTEXT:

- Business: {{company_name}} | Industry: {{industry}}
- Products/Services: {{products_and_services}}
- Target Customer: {{target_customer}}
- Unique Selling Points: {{usps}}
- Active Promotions: {{promotions}}
- Brand Tone: {{tone_of_voice}}
- Common Objections & Responses: {{objections_and_responses}}
- Booking Link: {{booking_link}}

BEHAVIOUR RULES:

1. Always open with the company name and a warm, personalised greeting.
2. Never reveal that you are an AI unless directly and explicitly asked. If asked, say "I'm the marketing assistant for {{company_name}} – here to help you."
3. Qualify every new inbound lead using the qualifying questions before discussing pricing.
4. If a lead is Hot (budget confirmed, timeline within 30 days, decision-maker), offer the booking link immediately.
5. If a lead is Warm, enrol them in the 3-step drip sequence.
6. If a lead is Cold, tag them and schedule a re-engagement in 14 days.
7. Handle every objection using the company's trained objection responses. Never concede on price without first presenting the USPs and value.
8. Keep messages concise: under 160 words for WhatsApp, conversational, never salesy.
9. Always end with a clear call to action.
10. Never make up product details, pricing, or availability. Only use information from the COMPANY CONTEXT.

1.5 Implementation Specification

Component	Specification
DB table: agent_mkt_leads	Columns: id, user_id, contact_id, qualification_score (1-10), tag (hot/warm/cold), stage, drip_step, last_contacted, booked_at, notes JSON
Drip sequence engine	A cron job runs every hour. For each warm lead WHERE last_contacted < NOW()-INTERVAL drip_delay HOUR AND drip_step < 4, send the next drip message and increment drip_step. Stop on reply.
Re-engagement cron	Daily at 8am: query contacts WHERE last_message_at < NOW()-14 DAYS AND tag != 'converted'. Send re-engagement message.
Upsell webhook handler	On WooCommerce/Shopify order event, query the product purchased, select the matching upsell script from agent_configs, wait 72 hours via queue, then send.
Lead scoring logic	Score 1-10: +3 budget confirmed, +2 timeline < 30 days, +2 decision maker, +1 replied within 1hr, +1 clicked link, +1 asked specific product question. Hot >= 7, Warm 4-6, Cold < 4.
Weekly report	Every Monday 8am cron: aggregate campaign_recipients + agent_mkt_leads for the week. Compose a summary message. Send via WhatsApp to the owner's registered number.
Booking detection	If message contains a booking link click event OR user says a time/date, extract intent and send confirmation + notify human sales team via WhatsApp.
Objection detection	Keyword matching on inbound messages: price expensive cost budget later not now think about trust reviews. On match, route to objection handling sub-flow.



AI Content Creator

On-demand and scheduled content production — blogs, captions, copy, scripts, campaigns

2.1 Human Role This Agent Replaces

A human content function employs a Copywriter who writes promotional and educational copy, a Content Strategist who plans what content to produce and when, a Blog/Article Writer, an Email/WhatsApp Campaign Writer, and a Brand Voice Guardian who ensures all content sounds consistent. This agent replaces all of them. It produces content on demand (when a user requests a piece) and on schedule (publishing a content calendar it builds itself), always in the company's brand voice.

2.2 Full Function List

Function	What the Agent Does Autonomously	Trigger
On-demand content generation	User sends a WhatsApp/Telegram message requesting a piece of content (e.g. 'write a promo for our 20% off sale'). Agent produces a finished draft immediately.	Inbound user request
Content calendar planning	On activation (and every month), the agent generates a 30-day content calendar with content types, topics, and publish dates based on the company's industry and goals.	Monthly cron + activation
WhatsApp campaign copy	Writes the full message copy for WhatsApp broadcast campaigns including headline, body, CTA, and emoji formatting — ready to send.	Scheduled or on-demand
Product/service descriptions	On request or when a new product is added (webhook), writes compelling descriptions in the company's tone for use in messages, posts, or web.	Webhook: new product or on-demand
FAQ content	Builds and maintains a knowledge base of FAQ answers in the company's voice, used by the Front Desk agent and the user's website.	Activation + on-demand updates
Blog / article outlines	Produces structured outlines (intro, H2 sections, conclusion, CTA) for blog posts on topics relevant to the company's industry.	On-demand
Testimonial / case study formatting	Takes raw customer feedback submitted by the user and formats it into a polished testimonial or short case study.	On-demand
Seasonal & event content	Automatically produces content for upcoming holidays, events, and seasonal peaks relevant to the company's market (e.g. Ramadan offers, Black Friday, New Year).	Scheduled 7 days before event
A/B message variants	For any campaign, produces two alternate versions of the message with different hooks and CTAs so the user can choose or test both.	On-demand or campaign creation

Content repurposing	Takes a long-form piece (blog, transcript, report) provided by the user and repurposes it into shorter WhatsApp messages, captions, and key-point summaries.	On-demand
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2.3 Onboarding Data Collected from Company

Data Field	Why the Agent Needs It
Business name, industry, and niche	Anchors all content to the right market and sector
Products / services (full list)	Enables product-specific copy, descriptions, and promos
Brand tone of voice (3 adjectives: e.g. bold, warm, professional)	Every piece of content must sound like the brand
Target audience description	Shapes language level, pain points addressed, and aspirations referenced
Content goals (leads / awareness / loyalty / education)	Determines content mix in the calendar
Competitors to differentiate from (optional)	Helps position content against market alternatives
Signature phrases or words to always use / avoid	Brand voice enforcement
Upcoming promotions or product launches	Pre-schedules related content
Preferred content types (WhatsApp, blog, short posts)	Shapes output formats
Industry keywords to include for discoverability	Ensures content is optimised for search where applicable

2.4 Master System Prompt

You are the AI Content Creator for {{company_name}}, a {{industry}} business.

Your role is to produce all written content this company needs – campaigns, product copy, blog outlines, WhatsApp messages, seasonal content, and FAQs – in a consistent brand voice.

You are the company's full content department.

COMPANY CONTEXT:

- Business: {{company_name}} | Industry: {{industry}} | Niche: {{niche}}
- Products/Services: {{products_and_services}}
- Target Audience: {{target_audience}}
- Brand Tone: {{tone_adjectives}} (e.g. bold, warm, professional)
- Content Goals: {{content_goals}}
- Signature Phrases: {{signature_phrases}}
- Words/Phrases to Avoid: {{avoid_phrases}}
- Upcoming Promotions: {{upcoming_promotions}}
- Preferred Content Formats: {{content_formats}}

BEHAVIOUR RULES:

1. Every piece of content must sound like it was written by a senior copywriter at {{company_name}}. Apply the brand tone in every sentence – not just the opening.
2. For WhatsApp messages: max 160 words, use line breaks generously, include 1-2 relevant emojis, end with a single clear CTA. Never write walls of text.
3. For blog outlines: produce H1 title, meta description (155 chars), 4-6 H2 sections with 2-sentence summaries each, and a CTA paragraph.
4. For product descriptions: open with the customer benefit (not the feature), use sensory language, end with urgency or social proof.
5. When producing a content calendar, output a table: Date | Content Type | Topic | Key Message | CTA. Cover 30 days. Mix promotional (40%), educational (35%), and engagement (25%) content.
6. Always produce A/B variants when the user asks for a campaign – label them Version A and Version B.
7. Never invent product specifications, pricing, or availability. Only use COMPANY CONTEXT data.
8. Never use jargon the target audience would not understand.
9. If the user provides raw material (a transcript, review, or report), acknowledge it and summarise it before repurposing into the requested format.

2.5 Implementation Specification

Component	Specification
DB table: agent_content_library	Columns: id, user_id, content_type (campaign/blog/faq/description/calendar), title, body TEXT, status (draft/approved/published), created_at, scheduled_at
Content calendar cron	On activation and 1st of each month: trigger the agent to produce a 30-day calendar. Store each day as a row in agent_content_library with scheduled_at set. Send the full calendar to the user via WhatsApp.
On-demand request handler	Inbound WhatsApp/Telegram message to the agent chat triggers the AI node. Detect content type from keywords (promo/blog/FAQ/description). Generate and return the content in the reply. Save to agent_content_library.
Seasonal event scheduler	Store a table of recurring events (Ramadan, Eid, Black Friday, New Year, Valentines, etc.) with lead times. 7 days before each event, trigger the agent to produce relevant content and send a draft to the user.
Product webhook	On new product webhook from WooCommerce/Shopify, extract product name, category, price. Trigger agent to write a product description and add to agent_content_library.
FAQ knowledge base	Store FAQ pairs in agent_faqs (user_id, question, answer). The Content Creator writes initial FAQs during onboarding. The Front Desk agent reads from this table for support responses.
Content approval flow	After content is generated and stored as 'draft', send it to the user with Approve / Edit buttons (WhatsApp quick replies). On Approve, status = 'approved'. Approved content is available to the Social Media agent for posting.



AI Social Media Manager

Content scheduling, community engagement, growth tracking — fully automated

3.1 Human Role This Agent Replaces

A human Social Media Manager handles content scheduling, caption writing, hashtag research, audience engagement (responding to comments and DMs), follower growth strategies, trend monitoring, competitor analysis, and weekly performance reporting — all manually across multiple platforms. This agent automates all of it. It pulls approved content from the Content Creator agent, schedules posts via social media APIs, monitors engagement, responds to inbound messages and comments, and delivers performance summaries directly to the business owner on WhatsApp.

3.2 Full Function List

Function	What the Agent Does Autonomously	Trigger
Post scheduling	Takes approved content from the Content Creator agent and schedules it to connected social platforms (Facebook, Instagram, LinkedIn, Telegram channel) at optimal times.	Content approved in library
Caption & hashtag writing	For each post, generates a platform-optimised caption with relevant hashtags (researched via AI), formatted for the specific platform.	Per post before scheduling
Inbound DM handling	Responds to direct messages received on connected social platforms — answers FAQs, qualifies leads, and routes complex queries to the Front Desk agent or human.	Inbound DM webhook
Comment monitoring & response	Detects comments on posts, responds to questions and compliments automatically, flags negative comments for human review with a WhatsApp alert.	Comment webhook / polling
Story / Reel content prompts	On a weekly basis, sends the user prompts and scripts for Stories or Reels they should record, tailored to the week's content theme.	Weekly cron — every Thursday
Trend monitoring	Monitors trending hashtags and topics in the company's industry. Alerts the user via WhatsApp when a trend is relevant and suggests a reactive post.	Daily cron — industry trend scan
Competitor monitoring	Tracks specified competitor accounts. Sends a weekly digest of what competitors posted, their engagement, and a suggested differentiation angle.	Weekly cron — Sunday
Audience engagement	Proactively engages with followers — likes relevant comments, responds to replies, and sends welcome messages to new followers on Telegram channel.	New follower / comment event
Growth reporting	Sends a weekly WhatsApp report: new followers, post reach, engagement rate, best-performing post, and recommended action for next week.	Every Monday 9am

Platform cross-posting	Repurposes approved WhatsApp broadcast content into social captions and posts it across all connected platforms automatically.	After broadcast send event
Negative sentiment escalation	Detects negative keywords in comments or DMs. Immediately alerts the human team via WhatsApp with the message and contact details for manual handling.	Real-time — sentiment scan

3.3 Onboarding Data Collected from Company

Data Field	Why the Agent Needs It
Connected social platforms (Facebook Page ID, Instagram, LinkedIn, Telegram channel)	Needed to authenticate and post via platform APIs
Posting frequency preference (e.g. daily, 3x/week)	Sets the scheduling cadence in the content calendar
Best posting times for your audience (or 'let the agent decide')	Optimises reach per post
Brand hashtag (your company's signature hashtag)	Included in every post
Industry hashtags (5-10)	Broadens post reach
Competitors to monitor (usernames/page names)	Powers weekly competitor digest
Topics to never post about	Brand safety guardrails
Tone for community replies (formal / casual)	Shapes DM and comment response style
Human escalation WhatsApp number	Where negative sentiment alerts and complex queries are sent
Telegram channel username (if applicable)	For Telegram-native posting and subscriber engagement

3.4 Master System Prompt

You are the AI Social Media Manager for {{company_name}}, a {{industry}} business.

You manage the company's full social media presence – scheduling posts, engaging with the audience, monitoring trends and competitors, responding to DMs and comments, and reporting on performance.

You work autonomously every day. You are the company's entire social media team.

COMPANY CONTEXT:

- Business: {{company_name}} | Industry: {{industry}}
- Connected Platforms: {{connected_platforms}}
- Posting Frequency: {{posting_frequency}} | Best Times: {{best_posting_times}}
- Brand Hashtag: {{brand_hashtag}} | Industry Hashtags: {{industry_hashtags}}
- Competitors to Monitor: {{competitors}}
- Community Reply Tone: {{community_tone}}

- Topics to Avoid: {{avoid_topics}}
- Escalation Contact: {{escalation_whatsapp}}

BEHAVIOUR RULES:

1. Every caption must be platform-native. Instagram: visual storytelling, emojis, 5-10 hashtags. LinkedIn: professional insight, no more than 3 hashtags. Telegram: direct, informational, link-focused. Facebook: conversational, community-driven.
2. When responding to DMs: acknowledge within the first sentence, answer the question using company knowledge, offer a next step (link, booking, product), and sign off warmly.
3. Never respond to a negative review or complaint with a defensive message. Always use: Acknowledge > Empathise > Offer resolution > Escalate if unresolved.
4. When a trend is detected: assess relevance score 1-10. Only alert the user if ≥ 7 . Provide a draft reactive post alongside the alert.
5. For competitor monitoring: never criticise competitors by name in public posts. Use insights only to position {{company_name}} more favourably.
6. Weekly report format: Followers gained | Total reach | Avg. engagement rate | Best post (link + metric) | Recommended focus next week.
7. Flag any comment containing: refund | complaint | scam | fake | disappointed | terrible – send immediately to escalation contact and do not reply publicly without human approval.

3.5 Implementation Specification

Component	Specification
DB table: agent_social_posts	Columns: id, user_id, content_library_id, platform, caption TEXT, hashtags, scheduled_at, published_at, platform_post_id, reach, likes, comments, status
Social API integrations	Facebook/Instagram: Meta Graph API (Pages + Instagram Basic Display). LinkedIn: LinkedIn Marketing API. Telegram: Bot API sendMessage to channel. Store OAuth tokens per user in agent_social_tokens.
Post scheduler	Cron every 30 minutes: query agent_social_posts WHERE status='scheduled' AND scheduled_at <= NOW(). For each, call the relevant platform API. On success, set status='published' and store platform_post_id.
DM webhook handler	Register webhooks on Facebook Messenger, Instagram DM, and LinkedIn messaging APIs. On inbound DM, route through the Social Media Agent AI node. Store conversation in a social_dm_threads table.
Comment polling	Every 15 minutes, use the Facebook/Instagram Graph API to fetch new comments on recent posts. Process each through sentiment analysis. Positive/neutral: generate reply. Negative: send WhatsApp alert to escalation contact.
Sentiment analysis	Use the AI node with a sentiment classification prompt: 'Classify this comment as POSITIVE, NEUTRAL, or NEGATIVE and identify if it contains a complaint or refund request.' Store result in comment record.
Trend monitor	Daily cron: call Twitter/X Trends API or Google Trends API for the company's industry keywords. Score relevance using the AI node. If score ≥ 7 , send WhatsApp alert to user with draft post.
Competitor monitor	Weekly Sunday cron: fetch recent posts from competitor page IDs via Facebook Graph API. Send a digest to the user with post summaries and engagement stats via WhatsApp.

Analytics aggregation	Weekly Monday cron: aggregate agent_social_posts metrics (reach, likes, comments) for the past 7 days. Compose and send the weekly report via WhatsApp.
Cross-post from broadcast	Listen for campaign_status = 'sent' events on campaigns table. Trigger agent to repurpose campaign body into a social caption and schedule across all platforms.



AI Front Desk & Support Agent

Receptionist, customer support, booking, complaint resolution — always on

4.1 Human Role This Agent Replaces

In a human organisation, the front desk and support function requires a Receptionist who greets and routes enquiries, a Customer Service Agent who answers questions and resolves issues, an Appointment Coordinator who manages bookings and reminders, a Complaints Handler who de-escalates and resolves problems, and an After-Sales Agent who follows up after a purchase or service. This agent performs all of these roles on WhatsApp and Telegram, 24 hours a day, handling every inbound message from a customer with the warmth, knowledge, and efficiency of a trained human team — escalating only when a human decision is genuinely required.

4.2 Full Function List

Function	What the Agent Does Autonomously	Trigger
First response & greeting	Instantly greets every new inbound contact by name (if available), identifies the nature of their enquiry, and routes them to the right sub-flow.	First inbound message from new contact
FAQ answering	Answers any question that matches the company's FAQ knowledge base instantly and accurately. Falls back to the AI's general knowledge of the industry if needed.	Any inbound question
Product / service enquiries	Provides detailed information about any product or service the company offers — pricing, availability, specifications, comparisons — pulled from onboarding data.	Inbound product enquiry
Appointment booking	Collects the customer's preferred date/time, checks against availability (via calendar API or simple slot list), confirms the booking, and sends a reminder 24 hours before.	Booking request keyword
Appointment reminders	Sends an automatic reminder 24 hours before and 1 hour before every booked appointment via WhatsApp.	Scheduled pre-appointment cron
Order status enquiries	Receives an order number from the customer, queries WooCommerce/Shopify via API, and returns real-time order status (processing, shipped, delivered).	'Order status' keyword or order number received
Complaint intake & de-escalation	Receives complaints with empathy and professionalism. Attempts to resolve using company refund/return policies. Logs the complaint. Escalates to human only if unresolved after 2 attempts.	Negative sentiment detected
Refund & return guidance	Walks the customer through the company's refund or return process step by step. Confirms receipt and logs the case. Notifies the human team with a summary.	Refund/return request

After-sales follow-up	Sends a follow-up message 48 hours after a completed service or delivery asking about satisfaction. If positive, requests a review. If negative, flags for human follow-up.	48h after order delivered webhook
Review / testimonial request	Sends a friendly review request with the company's Google / Trustpilot / Facebook review link after a positive interaction or purchase.	Positive sentiment detected or purchase confirmed
Human escalation	When a query is beyond the agent's scope, it tells the customer that a team member will follow up within X hours, logs the case, and sends a full conversation summary to the human team via WhatsApp.	Escalation trigger: 3 unresolved attempts, payment dispute, legal mention
Off-hours message	Outside business hours, informs the customer of business hours, collects their name and query, and queues a follow-up for the team the next morning.	Inbound message outside business hours
Language detection	Detects the language of the inbound message and responds in the same language automatically (using the AI model's multilingual capability).	Every inbound message

4.3 Onboarding Data Collected from Company

Data Field	Why the Agent Needs It
Business name, industry, location	Used in every greeting and response
Business hours (days + times + timezone)	Powers after-hours detection and messaging
Full product / service list with pricing and FAQs	The agent's primary knowledge base for answering enquiries
Refund and return policy (full text)	Used verbatim when handling refund/return requests
Booking / appointment slots or calendar link	Enables the booking flow
Human escalation WhatsApp number	Where escalation summaries and complaint alerts are sent
Response time promise (e.g. 'within 2 hours')	Used in off-hours messages and escalation notices
WooCommerce / Shopify API credentials (optional)	Enables live order status lookups
Review platform link (Google, Trustpilot, Facebook)	Used in review request messages
Complaint resolution authority (what can agent offer: discount, refund, replacement?)	Defines resolution limits before human escalation is required
Common complaints and preferred responses	Trains complaint de-escalation scripts

4.4 Master System Prompt

You are the AI Front Desk and Customer Support Agent for {{company_name}}, a {{industry}} business located in {{location}}. You are the first point of contact for every customer who messages this business

on WhatsApp or Telegram. You are warm, professional, efficient, and always on.

COMPANY CONTEXT:

- Business: {{company_name}} | Industry: {{industry}} | Location: {{location}}
- Business Hours: {{business_hours}} (timezone: {{timezone}})
- Products/Services & Pricing: {{products_and_services}}
- Refund & Return Policy: {{refund_policy}}
- Booking Slots / Calendar Link: {{booking_info}}
- Escalation Contact: {{escalation_whatsapp}}
- Response Time Promise: {{response_time}}
- Review Link: {{review_link}}
- Resolution Authority: {{resolution_authority}}
- Common Complaints & Responses: {{complaint_scripts}}

BEHAVIOUR RULES:

1. Greet every new contact by name if available. Always open with the company name and a warm welcome.
Example: "Hi [Name]! Welcome to {{company_name}}. How can I help you today?"

2. Respond in the same language the customer writes in. If uncertain, default to English.

3. Never make the customer feel like they are talking to a bot. Be conversational, use their name, and acknowledge their feelings before providing information.

4. For complaints: Lead with empathy ('I completely understand your frustration and I'm sorry to hear this').

Never be defensive. Offer a resolution within your authority. If unresolved after 2 attempts, escalate.

5. For refund/return requests: follow the policy exactly. Do not deviate. If the request is outside policy,

say 'Let me get a team member to help you with this' and escalate.

6. For bookings: confirm the date, time, and service clearly. Send a summary of the booking in your reply.

Set a reminder job for 24 hours before the appointment.

7. Never discuss competitor businesses. If asked, say 'I can only speak to what we offer at {{company_name}}.'

8. Outside business hours: 'We're currently closed – our hours are {{business_hours}}.

I've noted your query and a team member will follow up by {{next_business_day}}. Can I take your name?'

9. Escalation message format (sent to human team): ESCALATION ALERT | Customer: [name] | Query: [summary]

| Attempts: [n] | Conversation: [last 3 messages] | Action needed: [what you tried]

10. After a positive resolution, always ask: 'We'd love to hear your feedback!

Would you mind leaving us a quick review? [{{review_link}}]'

4.5 Implementation Specification

Component	Specification
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DB table: agent_support_cases	Columns: id, user_id, contact_id, case_type (enquiry/complaint/refund/booking/escalation), status (open/resolved/escalated), attempts INT, summary TEXT, created_at, resolved_at
DB table: agent_bookings	Columns: id, user_id, contact_id, service, date, time, status (pending/confirmed/reminded/completed/cancelled), reminder_sent_at
DB table: agent_faqs	Shared with Content Creator. Columns: id, user_id, question, answer, category, created_at. Agent performs fuzzy match on inbound questions.
Business hours detection	Every inbound message passes through a middleware that checks current UTC time against the stored business_hours JSON for the user. If outside hours, route to off-hours sub-flow.
Order status lookup	On receiving an order number, call WooCommerce REST API or Shopify Admin API with stored user credentials. Return status, items, and tracking link. Cache result for 5 minutes per order number.
Booking flow	Collect: service > preferred date > preferred time. Check agent_bookings for conflicts. Confirm and insert. Queue two reminder jobs in Bull: -24h and -1h from appointment datetime.
Escalation trigger	After 2 unresolved AI responses on the same case (attempts >= 2), or on detection of escalation keywords (legal lawyer sue report fraud police refund refused), automatically: update case status to 'escalated', send escalation WhatsApp to human team, tell customer a team member will follow up.
Sentiment monitor	Every inbound message passes through a quick AI sentiment check (POSITIVE/NEUTRAL/NEGATIVE). NEGATIVE with complaint keywords opens a new case or appends to existing. POSITIVE after resolution triggers review request.
After-sales follow-up	On order delivered webhook: insert a delayed job (Bull, delay = 48 hours). On execution, send satisfaction message. Parse reply for sentiment. Positive: send review link. Negative: open escalation case and alert human.
FAQ matching	Use AI embedding similarity or simple keyword matching on the agent_faqs table. If confidence > 0.8, return the stored answer. If < 0.8, use the AI node with company context to generate a response.
Language detection	First message from a new contact: AI node classifies the language. Store in contact record as preferred_language. All subsequent prompts prepend: 'Respond in [language].'

Shared Technical Specification — All Agents

Database Tables (Shared)

Table	Key Columns	Purpose
agent_configs	id, user_id, agent_type (marketing/content/social/support), system_prompt TEXT, is_active, ai_model, created_at	Master config per activated agent per user
agent_company_data	id, user_id, agent_type, field_key VARCHAR, field_value TEXT, updated_at	Company onboarding answers that populate the system prompt COMPANY CONTEXT block
agent_activity_log	id, user_id, agent_type, action, details JSON, contact_id, status, created_at	Full audit trail of every action each agent takes
agent_schedules	id, user_id, agent_type, schedule_type (cron/delay), cron_expression, next_run_at, last_run_at, payload JSON, status	All scheduled jobs across all agents
agent_faqs	id, user_id, question TEXT, answer TEXT, category, embedding VECTOR(1536)	Shared FAQ knowledge base (written by Content Creator, read by Support Agent)
agent_social_tokens	id, user_id, platform, access_token (encrypted), refresh_token (encrypted), expires_at, page_id	OAuth tokens for social platform APIs

Activation Flow (from FLYN Dashboard)

Step	What Happens	Implementation
1. User clicks 'Activate' on agent addon	Agent record created in agent_configs with is_active=false. Onboarding conversation initiated via WhatsApp/Telegram.	INSERT into agent_configs. Trigger onboarding flow for agent_type.
2. Onboarding wizard runs	Agent sends 8-12 questions via WhatsApp. Answers stored in agent_company_data as field_key/value pairs.	Flow Builder onboarding sub-flow. Take Input nodes. Store each answer.
3. System prompt assembly	System assembles the master prompt by injecting all agent_company_data values into the prompt template for that agent_type.	Server-side template engine: replace {{field_key}} with field_value. Store result in agent_configs.system_prompt.
4. Agent goes live	agent_configs.is_active = true. Cron schedules registered. Inbound message trigger activated. Confirmation sent to user.	Update agent_configs. Register cron jobs in agent_schedules. Confirm via WhatsApp.
5. User can adjust settings	From the FLYN dashboard, the user can re-run onboarding for any field, pause the agent, or change the AI model.	Dashboard Settings panel. Updates agent_company_data and regenerates system_prompt.

AI Model & Prompt Injection Architecture

Each agent uses the FLYN AI Chat node (OpenAI GPT-4o by default; switchable to Gemini 1.5 Pro or DeepSeek V3 per user preference). The AI node receives: (1) the master system prompt from agent_configs.system_prompt with all COMPANY CONTEXT fields already injected, (2) the last 10 messages from the current conversation

thread as the conversation history, and (3) the current inbound message as the user turn. The model returns the agent's reply. For scheduled/cron actions, the node receives the system prompt + the task instruction (e.g. 'Generate a 30-day content calendar') with no user turn.

```
// Standard AI Node payload for all agents
{
  "model": "gpt-4o", // or gemini-1.5-pro / deepseek-v3 per user setting
  "max_tokens": 1000,
  "messages": [
    { "role": "system", "content": "" },
    { "role": "user", "content": "" },
    { "role": "assistant", "content": "" },
    { "role": "user", "content": "" },
    { "role": "assistant", "content": "" },
    ...up to 10 prior turns...
    { "role": "user", "content": "" }
  ]
}
```

Dashboard UI — Agent Management Panel

UI Element	Function	Implementation
Agent Addon Cards	4 cards on the addons page: Marketing, Content, Social, Support. Each shows status (Active/Inactive), last action, and a toggle.	Query agent_configs for user. Render status badge. Toggle calls PATCH /agents/:type/toggle.
Activate button	Triggers onboarding wizard via WhatsApp/Telegram. Disabled until a channel is connected.	POST /agents/:type/activate. Checks active channel. Fires onboarding flow.
Settings panel	For each active agent: re-run onboarding, update individual fields, change AI model, pause/resume.	GET/PATCH /agents/:type/settings. Renders field_key/value pairs from agent_company_data.
Activity log	Shows the last 50 actions the agent took with timestamps and outcomes.	GET /agents/:type/activity. Queries agent_activity_log for user + agent_type.
Performance metrics	Per-agent KPI cards: Marketing (leads captured, conversions); Content (pieces created); Social (posts published, engagement); Support (cases resolved, avg response time).	Aggregate queries per agent type. Cache for 1 hour.
Manual override	User can send a one-off message or instruction to any agent via a chat input in the dashboard.	POST /agents/:type/instruct. Fires a one-time AI node call with the instruction as the user turn.

This document defines the complete specification for the four FLYN AI Smart Agent Addons. All agents are built on the existing FLYN flow engine and AI Chat node infrastructure. No third-party automation platforms are required. The onboarding wizard, system prompt assembly, cron scheduling, and dashboard UI are the four new engineering deliverables. Prepared: 01 April 2026 — FLYN AI Internal.